



OPEN EEG-based emotion recognition using multi-scale dynamic CNN and gated transformer

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Emotions play a crucial role in human thoughts, cognitive processes, and decision-making. EEG has become a widely utilized tool in emotion recognition due to its high temporal resolution, real-time monitoring capabilities, portability, and cost-effectiveness. In this paper, we propose a novel end-to-end emotion recognition method from EEG signals, called MSDCGTNet, which is based on the Multi-Scale Dynamic 1D CNN and the Gated Transformer. First, the Multi-Scale Dynamic CNN is used to extract complex spatial and spectral features from raw EEG signals, which not only avoids information loss but also reduces computational costs associated with the time-frequency conversion of signals. Then, the Gated Transformer Encoder is utilized to capture global dependencies of EEG signals. This encoder focuses on specific regions of the input sequence while reducing computational resources through parallel processing with the improved multi-head self-attention mechanisms. Third, the Temporal Convolution Network is used to extract temporal features from the EEG signals. Finally, the extracted abstract features are fed into a classification module for emotion recognition. The proposed method was evaluated on three publicly available datasets: DEAP, SEED, and SEED_IV. Experimental results demonstrate the high accuracy and efficiency of the proposed method for emotion recognition. This approach proves to be robust and suitable for various practical applications. By addressing challenges posed by existing methods, the proposed method provides a valuable and effective solution for the field of Brain-Computer Interface (BCI).

Keywords EEG signals, Temporal-spatial features, Multi-scale dynamic 1D CNN, Gated transformer encoder, Temporal convolution network

Emotions are physiological and cognitive reactions that humans experience in response to events or situations, which play a crucial role in human thoughts, cognitive processes, and decision making. Emotion analysis has found wide applications in various fields¹, including disease diagnosis, psychological analysis and evaluation, and social behavior analysis², etc. Traditional emotion analysis methods used non-physiological signals to detect the emotion state, including facial expression, speech, and behavioral characteristics^{3–6}. However, these methods are often influenced by individual and cultural differences, which leads to subjective interpretations of emotions and potential misidentification. With the rapid development of brain-computer interface (BCI)^{7–9}, Electroencephalography (EEG)¹⁰ is a technique that captures brain signals corresponding to various states from the scalp surface area through the use of BCI. These signals are often categorized into different frequency bands, each associated with specific brain activities. These frequency bands are delta, theta, alpha, beta, and gamma, which covers a range from 0.1 Hz to more than 100 Hz. EEG has been extensively used in emotion research because of its high temporal resolution, real-time monitoring, portability, and cost-effectiveness. Therefore, the application of EEG to emotion analysis has garnered widespread attention and has become a research hotspot.

Many emotion recognition methods based on EEG signals have been proposed. The clustering method, as described in^{11,12}, organizes EEG signals into clusters with similar characteristics based on the intrinsic structure of the data. While this method is known for its simplicity and efficiency, it is susceptible to producing incorrect detections. The primary challenges lie in the selection of initial centers and determining the appropriate number of clusters. Errors in these aspects can lead to suboptimal clustering results and compromise the accuracy of the detection process. Machine learning methods, as outlined in^{13–16}, employ manual feature selection and classifiers for emotion detection, yielding improved performance compared to clustering methods. The key advantage lies

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in the machine learning model's ability to learn the relationships between input features and corresponding labels during the training process. This enables the model to generalize effectively when confronted with new data. However, it's worth noting that the use of artificial feature extraction to construct EEG signal features in these methods can potentially limit their generalization ability. With the development of deep learning, Convolutional Neural Network (CNN)^{17–19} has been utilized for EEG emotion detection. In this approach, EEG signals are first converted into spectrum images by time-frequency conversion, and then these spectrum images are fed into the CNN network for feature extraction and classification. Compared to traditional machine learning, CNN demonstrates superior performance due to its enhanced generalization ability. Dynamical graph convolutional neural networks (DGCNN)²⁰ to classify graphs encoded with spatiotemporal information of the emotion EEG source, and it can effectively extract spatiotemporal features. However, one limitation of them is its inability to capture temporal features since EEG is a time serial signal, which leads to potential shortcomings in detection accuracy. Given that LSTM or RNN can capture temporal characteristics, they are introduced in combination with CNN for emotion detection^{21–23}. CNN is employed to extract local feature of EEG signal, while LSTM or RNN extract temporal features. This approach tends to yield more satisfactory results compared to using CNN alone, but these methods require preprocessing of the data using time-frequency transformation, which could potentially lead to a loss of information and increase computational demands. The Transformer²⁴ is an Encoder-Decoder architecture with a Self-Attention Mechanism introduced by Vaswani et al. in 2017. Initially proposed for natural language²⁵ processing tasks like machine translation, its versatility has led to widespread adoption in various domains, including computer vision, speech recognition, and more. The advantages of the transformer are attributed to its parallelization capabilities, scalability to handle long sequences, and its effectiveness in capturing contextual information. Consequently, the transformer framework has been increasingly applied to EEG emotion detection^{26,27}, which is achieving better performance than the combined method of CNN and LSTM.

The above methods have achieved satisfactory results, but the following challenges still need to be addressed: (1) Reduction of complexity. The currently existing methods require a pre-processing process, that is, time-frequency transformation, which can be effective in extracting features from signals in both time and frequency domains. However, this approach poses the following problems: firstly, time-frequency conversion results in information loss and increased computational load; secondly, the preprocessing introduces complexity into the system. (2) Prediction accuracy and robustness. Given that EEG signals are the time sequence data, different segments of the sequence have varying levels of importance and influence on the prediction result. Existing models often overlook this issue, which can result in failure to capture crucial patterns or information present in the EEG data. This can ultimately lead to a reduction in both performance and generalization capabilities. In this paper, motivated by the practical application of EEG emotion detection, we propose a novel end-to-end emotion recognition method from EEG signals using a Multi-Scale Dynamic CNN and a Gated Transformer Encoder, named MSDCGTNet. Experiments were conducted on three public datasets: DEAP, SEED, and SEED_IV, achieving impressive accuracy results of 99.66%, 98.85%, and 99.67%, respectively. Our model not only provides novel and robust tools for EEG-based emotion recognition but also holds significant potential for diverse applications within brain-computer interface (BCI) systems.

In short, the novelty and significant contributions of the study are as follows:

- (1) In this paper, we propose a novel end-to-end framework that utilizes the Multi-Scale Dynamic 1D CNN and the Gated Transformer Encoder for EEG-based emotion recognition. This unique framework addresses the challenges of EEG feature extraction and emotion classification as a unified process. Additionally, it reduces the model's complexity.
- (2) The Multi-Scale Dynamic 1D CNN is employed to extract spatial-spectral features from raw EEG signals, which not only helps avoid information loss caused by time-frequency conversion but also significantly enhances recognition performance.
- (3) The Gated Transformer Encoder is used to extract global features from EEG signals. The gated layers can learn to dynamically select relevant features, which is particularly useful for EEG data where important signals may vary across time and channels. By incorporating non-linear transformations, the model can better capture complex relationships in the data that are not easily addressed by linear operations alone. Furthermore, the improved multi-head self-attention mechanism efficiently handles long EEG sequences compared to traditional Transformers, thereby reducing computational complexity and making it suitable for real-time processing of large-scale EEG signals.
- (4) We compared our method with other state-of-the-art approaches using three public datasets: DEAP, SEED, and SEED_IV. The results demonstrate that our method achieves superior recognition accuracy and efficiency compared to other leading methods.

The remaining sections of this article are as follows: In “[Methods](#)” the detailed structure of our method is reviewed. In “[Experiment](#)”, the experimental dataset, experimental evaluation metrics, performance comparison of the method with other methods are described and future direction is discussed. In “[Conclusion](#)”, the conclusion is presented.

Methods

As shown in the Fig. 1, the method based on Multi-Scale dynamic 1DCNN and the Gated transformer for emotion recognition from raw EEG data is introduced in this paper. The method is divided into four parts, (1) Multi- Scale dynamic 1DCNN, which extracts the local characteristic from the raw EEG data. (2) The Gated Transformer Encoder, which extracts the global features. (3) Temporal Convolutional Network (TCN), which extract discriminative time series features while integrating historical information of the time series. (4)

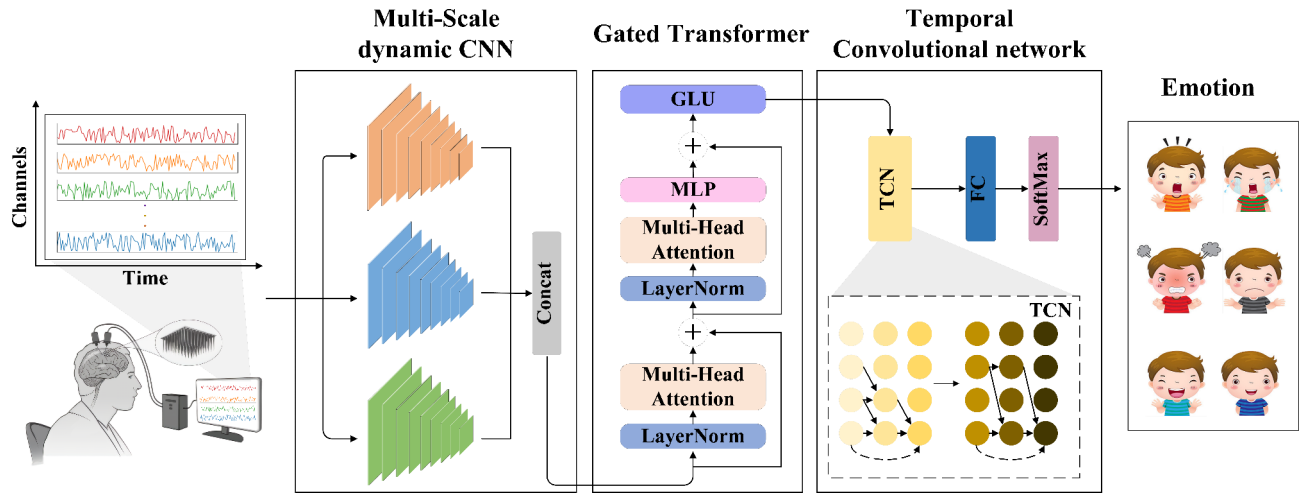


Fig. 1. Structural flowchart of the MSDCGTNet model.

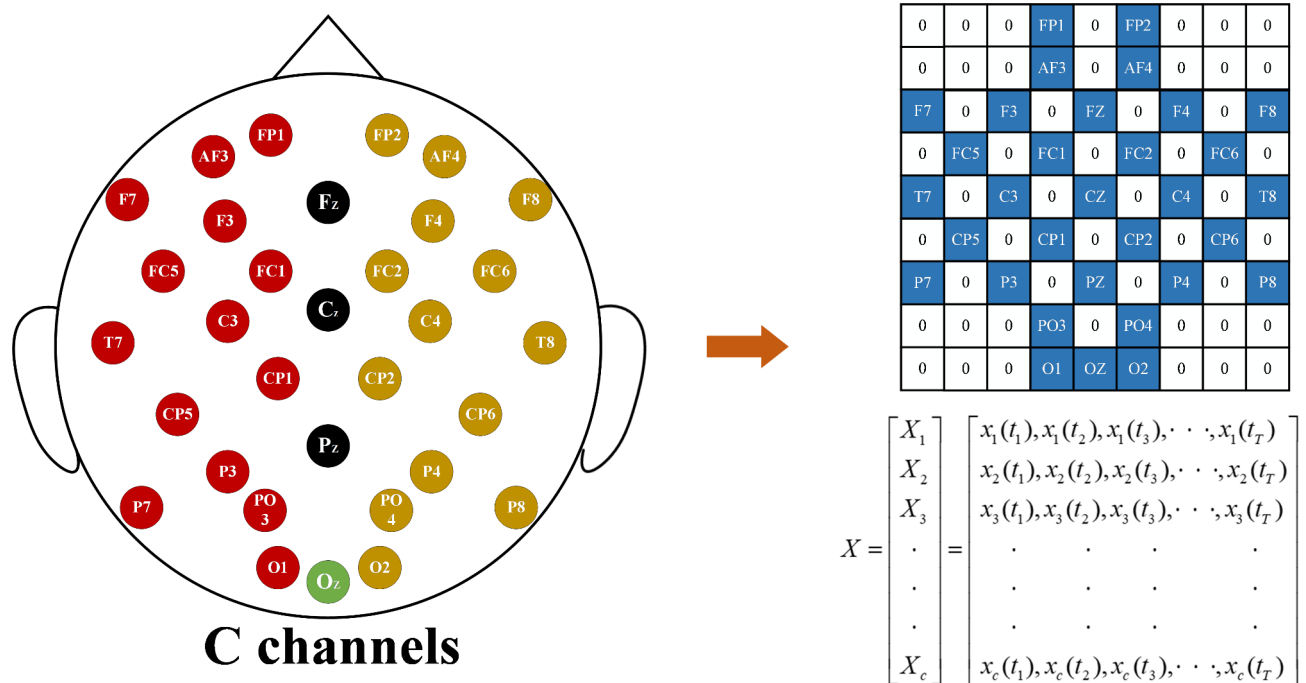


Fig. 2. The electrode map of the data collection channel.

SoftMax, which selects the class with the highest probability as the final prediction. This section details the main idea of MSDCGTNet.

EEG signal description

EEG data can be regarded as two-dimensional sequence data with spatial dimensions (EEG electrode channels) and temporal dimensions (data sampling points). The spatial dimension reflects the activation state of the brain in different functional areas, while the temporal dimension reflects the temporal change of brain activity. Since EEG segments of varying lengths contain different emotion information, we use time windows to segment the EEG signals. Each EEG segment signal is captured by C channels over a time interval with T timestamps. This data is then represented as a $C \times T$ matrix, as shown in Fig. 2.

Where $X_i = [x_i(t_1), x_i(t_2), x_i(t_3), \dots, x_i(t_T)]$ is the i -th channel over a time interval.

The objective of the MSDCGTNet model is to map the input EEG data X_i to its corresponding category y_i . Given a set of labeled EEG data $U = \{(X_i, y_i)\}_{i=1}^m$, where $y_i \in \{1, \dots, n\}$ is the category label corresponding to trial X_i , n is the total number of categories defined by set U , and m is the total number of samples.

Multi-scale dynamic 1D CNN

At present, most existing methods for EEG-based emotion recognition rely on time-frequency conversion since CNN network typically requires image-like inputs. This involves transforming EEG signal into various spectrograms, such as Fbank, MFCC, and speech spectrograms. From which multiple features are then extracted for recognition. However, these methods have several limitations: (1) Information Loss. Time-frequency conversion leads to information loss, thereby reducing the accuracy of recognition. (2) Computational Time. The time-frequency conversion process consumes significant computing resources, which can hinder real-time EEG processing. (3) Non-stationary and Dynamic. EEG signals are non-stationary and dynamic, and features extracted through serial convolution may be too simplistic, decreasing the model's generalization ability. To address these limitations, we propose the Multi-Scale Dynamic 1D CNN, inspired by the pioneering work discussed in²⁶. This method allows for efficient feature extraction directly from raw EEG signals, circumventing the drawbacks associated with time-frequency conversion.

As shown in Fig. 3, the multi-scale dynamic convolution module is designed to effectively capture the dynamic frequency characteristics of EEG signals by using convolution kernels of different sizes. The detailed steps are as follows: ① Temporal Convolution. Each convolution kernel applies a frequency filter for temporal convolution to capture variations in the time domain. ② Spatial Convolution. Convolution with a kernel size of (C, 1) is performed to process EEG signals in the spatial domain, focusing on the spatial relationships between EEG channels. ③ Dimensionality Reduction. An average pooling layer with a size of (1, 4) is used to reduce the dimensionality of the temporal features, and dropout technology is employed to prevent overfitting. ④ Feature Fusion. The concatenation layer fuses the output features from the pooling layer. As a result, the tensor obtained after the multi-scale dynamic convolutional module represents signal sources from different frequencies and electrode channels.

In the first layer of multi-scale dynamic convolution, the temporal convolution kernels are sized according to a specific ratio of the EEG sampling frequency f . The α^i determines the relationship between the convolution kernel size K and the EEG sampling frequency f , where i represents the scale level in the multi-scale network. The network has multiple scales L , and for each scale i (ranging from 1 to L), α is set to 0.5, the kernel size is adjusted according to the radio coefficient. Let represents the size of the i -th level convolution kernel. The size is determined by the following equation:

$$k_T^i = (1, \alpha^i f), i \in [3, 4, 5] \quad (1)$$

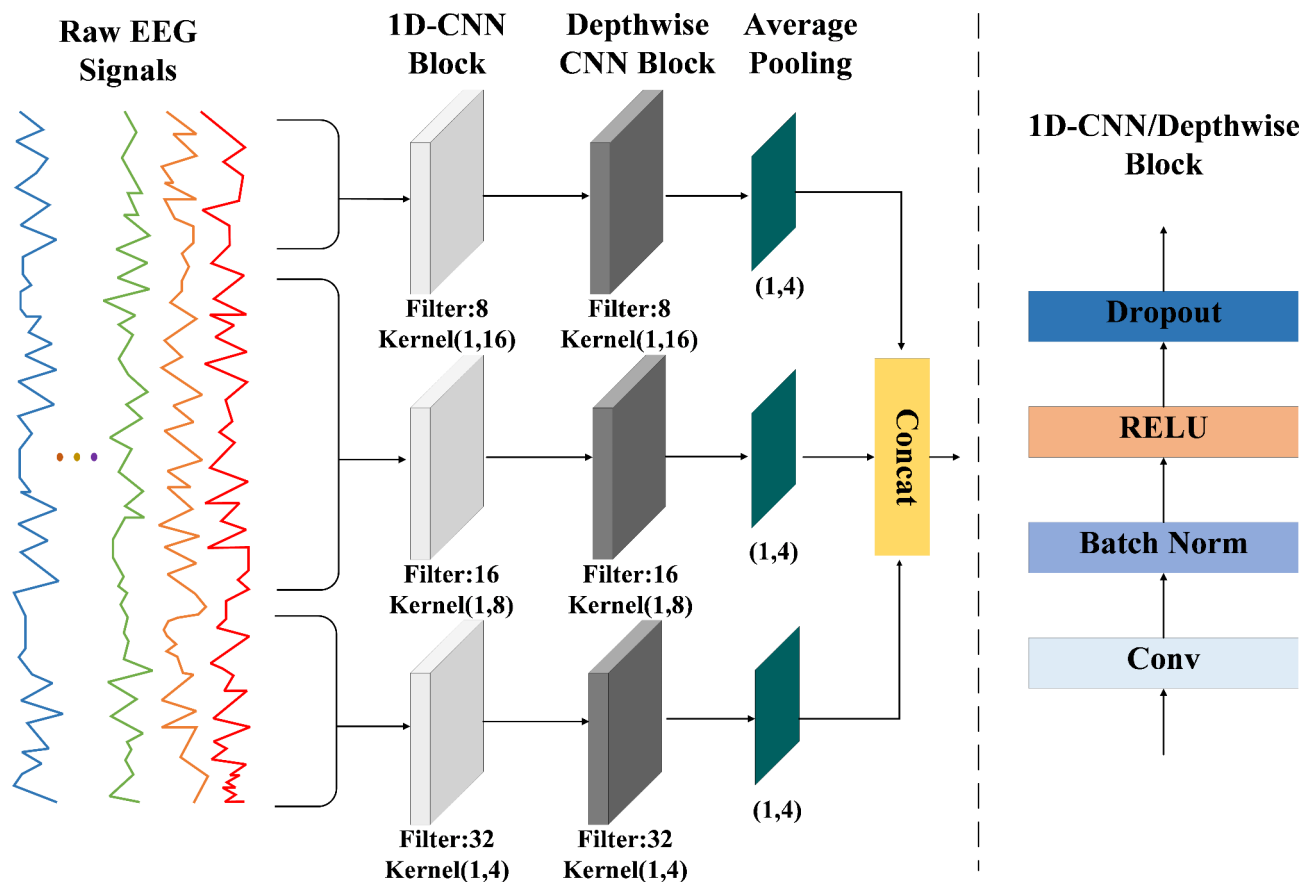


Fig. 3. The detailed structure of multi-scale dynamic CNN.

Considering that excessive convolution operations can increase the number of parameters, and the brain signals related to emotions are in the following frequency bands²⁸: α : 8–13 Hz, β : 14–30 Hz, γ : 31–51 Hz, thereby affecting operational efficiency, the number of scale level L is set to 3. For i ranging from 3 to 5, the ratio coefficients become [0.125, 0.0625, 0.03125], meaning the sizes of correspond to [(1, 16), (1, 8), (1, 4)]. This adjustment ensures that the model maintains its classification efficiency without compromising on performance. From the frequency perspective, when the length of k is set to different ratios of f , the temporal convolution layers can capture frequency information of 8 Hz, 16 Hz, and 32 Hz, respectively²⁹. From the temporal perspective, these convolution kernel sizes correspond to time steps of 125 ms, 62.5 ms, and 31.25 ms in duration.

The gated transformer encoder

Each convolutional layer under the CNN framework shares weights, which means that they focus equally on each part of the EEG signal. To learn the global dependencies in EEG signals and focus on the relevant parts crucial for classification tasks, the Gated Transformer Encoder is proposed to capture global features in EEG signals, which are crucial for understanding the overall context and dependencies within the data. The multi-head self-attention mechanism in the Transformer encoder allows the model to focus on different parts of the input data simultaneously, identifying the most relevant features for the classification task. The GLU layer combines linear transformations with a gating mechanism, which helps in selectively passing important features while filtering out less relevant information. This improves model performance and efficiency by reducing computational overhead.

The output of the Multi-Scale Dynamic Convolution module is defined as $O = \{x_1, x_2, x_3, \dots, x_n\}$, which serves as the input to the Gated Transformer Encoder for further processing and feature extraction. Here, O consists of N vectors, each of which has a dimension d . The Gated Transformer Encoder proposed in this paper incorporates a gated linear unit (GLU) layer to the traditional Transformer encoder. It is composed of l identical layers, each consisting of three modules: the multi-head attention mechanism (MHA), the multi-layer perceptron (MLP), and the gated linear unit (GLU). The Gated Transformer Encoder can be expressed in formulas (2) to (4).

$$O'_l = MHA(LayerNorm(O_{l-1})) + O_{l-1} \quad (2)$$

$$O'_l = MLP(LayerNorm(O'_l)) + O'' \quad (3)$$

$$O_l = GLU(O'_l) \quad (4)$$

The improved multi-head self-attention mechanism

Since Artificial Neural Networks (ANN) tend to lose a significant amount of information during long-distance propagation and are not very sensitive to feature importance, they can struggle to capture complex dependencies in data, which can lead to suboptimal performance in tasks that require understanding intricate relationships or maintaining long-range contextual information. This limitation often necessitates the use of more advanced architectures, such as Transformers, which utilize self-attention mechanisms to better preserve and emphasize critical features across long sequences. However, the computational complexity of the Multi-Head Self-Attention mechanism is dominated by the dot-product calculation, which is $O(n^2 \cdot d)$. This quadratic dependence on the sequence length n makes self-attention computationally expensive for long sequences. To address the computational and memory challenges associated with self-attention, an improved attention mechanism method³⁰ is proposed to reduce complexity by projecting the keys and values to a lower-dimensional space, which is achieve linear complexity relative to sequence length. This approach aims to retain the benefits of self-attention while making it feasible to apply to longer sequences and larger datasets.

As shown in Fig. 4, the key idea behind the improved method is to replace the standard self-attention mechanism in transformers with a more efficient linear attention mechanism. The detailed steps are as follows:

(1) Linear projection of keys and values.

In the standard Transformer attention mechanism, Q , K , and V are the queries, keys, and values, respectively. Instead of directly computing K and V , the improved method introduces two learned projection matrices $P_K \in \mathbb{R}^{n \times k}$ and $P_V \in \mathbb{R}^{n \times k}$, which project the original keys K and values V into a lower-dimensional space of dimension, where $k < n$ (sequence length). The formula of projected Keys K and Values V is as follows.

$$\begin{aligned} K_p &= K \cdot P_K \\ V_p &= V \cdot P_V \end{aligned} \quad (5)$$

These matrices are used to reduce the dimensionality of the keys and value from $\mathbb{R}^{n \times d}$ to $\mathbb{R}^{k \times d}$.

(2) Query, key, and value computation.

Compute the query (Q), key (K), and value (V) matrices using learned weight matrices:

$$Q = X \cdot W_Q, K = X \cdot W_K, V = X \cdot W_V \quad (6)$$

(3) Scaled dot-product attention.

Compute attention scores using the scaled dot-product attention with projected keys:

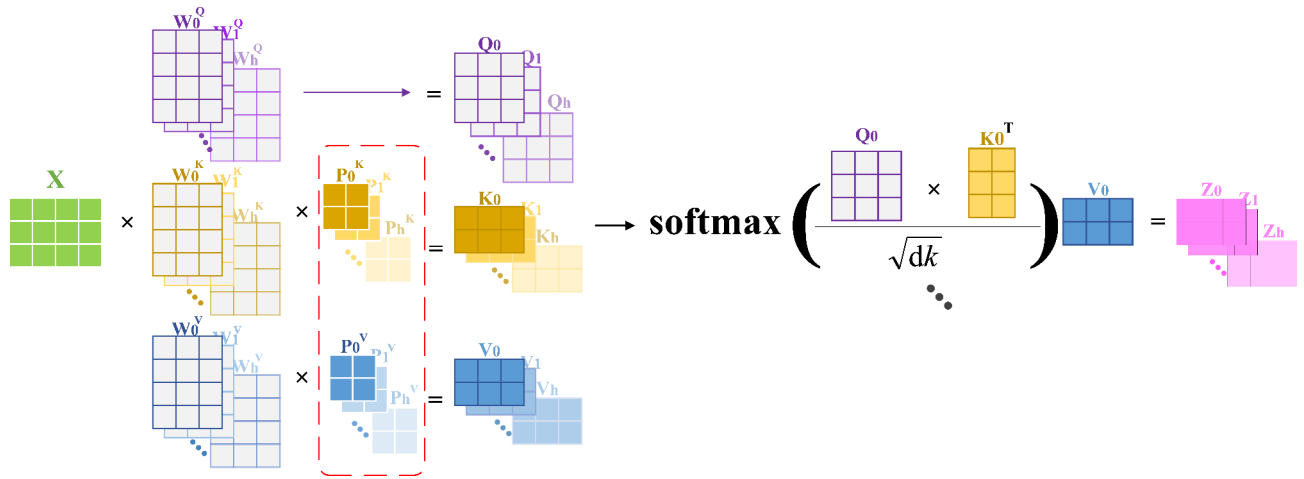


Fig. 4. The detailed structure of the improved multi-head self-attention mechanism.

$$Attention(Q, K_p, V_p) = softmax\left(\frac{Q \cdot K_p^T}{\sqrt{d_k}}\right) \cdot V_p \quad (7)$$

The softmax function normalizes the attention scores, ensuring they sum to 1.

(4) Multi-head attention.

Similar to the standard Transformer, the improved can utilize multiple attention heads. Each head operates independently on different parts of the input space:

$$head_i = Attention(QW_{Q,i}, K_pW_{K,i}, V_pW_{V,i}) \quad (8)$$

Concatenate the outputs of all heads and apply a final linear transformation:

$$MultiHead(Q, K_p, V_p) = Concat(head_1, head_2, \dots, head_h) \cdot W_O \quad (9)$$

$W_{Q,p}, W_{K,p}, W_{V,p}, W_O$ are learned matrices for each head, and h is the number of attention heads.

(5) Feedforward network.

Pass the result through a position-wise feedforward neural network, which consists of two linear transformations with a ReLU activation in between:

$$FFN(x) = ReLU(x \cdot W_1 + b_1) \cdot W_2 + b_2 \quad (10)$$

This network is applied independently to each position in the sequence.

(6) Residual connections and layer normalization.

Add residual connections around each sub-layer followed by layer normalization:

$$LayerNorm(x + SubLayer(x)) \quad (11)$$

This helps stabilize training and allows for better gradient flow through the network.

The gated linear unit (GLU)

As show in Fig. 5, to retain the temporal and spatial information when processing EEG data and to speed up operations during parallel data processing in the Transformer, the Gated Linear Unit (GLU)³¹ is integrated into the Transformer encoder. The GLU controls the information transmission of EEG signals during training, thereby improving the network's learning performance. The GLU consists of a convolutional layer and a gating layer, and its calculation is shown in Equation:

$$H(x) = (x * W + b) \otimes \sigma(x * U + c) \quad (12)$$

where x is the embedding vector; W and U are the convolution kernel in the convolution operation, b and c are the bias parameters, $*$ is the convolution operation, $\otimes$ is the multiplication of the corresponding elements of the matrix, σ is the sigmoid activation function. The sigmoid activation function, denoted as σ , compresses

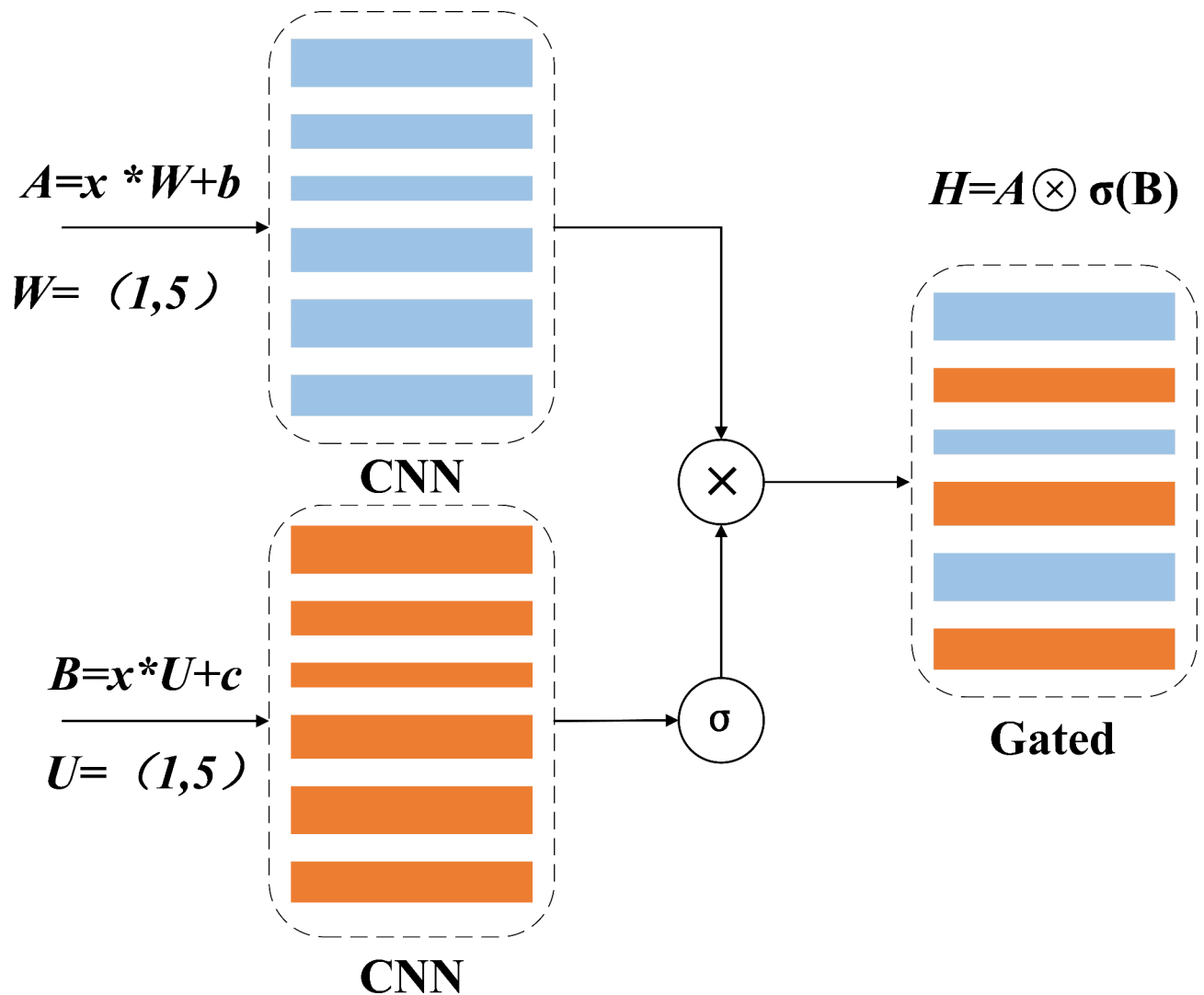


Fig. 5. The detailed structure of GLU.

all information before the current moment in the time window. This facilitates the control of effective EEG information entering the next layer.

Temporal convolutional network (TCN)

After extracting EEG signal features of different frequencies and electrode channels, TCN module³² is applied to extract discriminative time series features while integrating historical information of the time series. The TCN is composed of stacked residual blocks, with each residual block consisting of two dilated causal convolution (DCC) layers with different dilation coefficients, as shown in Fig. 6. Each convolution layer is followed by batch normalization (Batch Norm) and an Exponential Linear Unit (ELU) activation function.

Classification

Emotion recognition from EEG signal is actually a multi-classification problem. SoftMax is often used in the final layer of a neural network for multi-class classification tasks. The predicted class is the one with the highest probability according to the SoftMax output. The SoftMax function is differentiable, which is crucial for training neural networks using gradient-based optimization algorithms. Therefore, SoftMax is employed to emotion classification in this paper.

Given an input vector x of length M , the SoftMax function computes a vector $S(x)_i$ of the same length M . The SoftMax function is defined as follows:

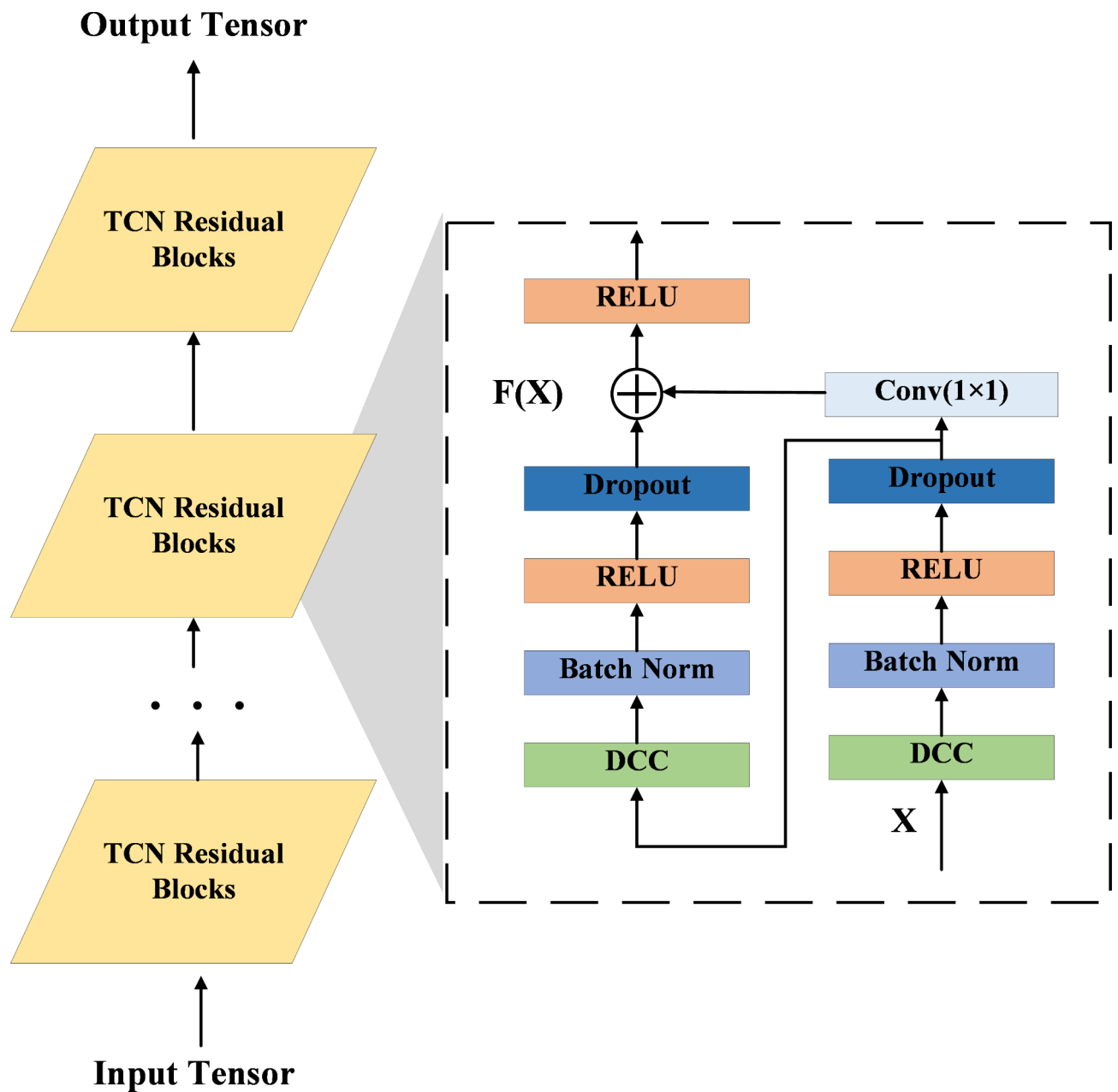


Fig. 6. The detailed structure of TCN.

$$S(x)_i = \frac{e^{x_i}}{\sum_{j=1}^M e^{x_j}} \quad (13)$$

$$j = \arg \max_{i=1}^M P(S_i) \quad (14)$$

where e is the base of the natural logarithm (Euler's number), x_i is the i -th element of the input vector x , $S(x)_i$ is the i -th element of the output vector $S(x)$, M is the total number of classes. The $\arg \max$ designates the selection of the category label that yields the maximum conditional probability. Therefore, the category with the highest probability j is selected as the prediction result.

Experiment

In this section, experimental results are presented to verify the performance of our method. First, the experiment data, environment, evaluation metrics and hyperparameter setting are introduced in detail. Then, both the efficiency and the effectiveness of our method are evaluated and compared with several state-of-the-art methodologies. Finally, the ablation experiments and parameter analysis were employed to further evaluate our proposed approach.

Datasets

Three open-source datasets (DEAP, SEED, SEED_IV) are utilized as the test datasets to access the performance of the proposed method. These datasets used in our experimentation and the experimental protocol are described in detail as follows.

- (1) The DEAP dataset³³, created by the Queen Mary University of London, is designed for emotion analysis. It consists of physiological signals recorded from 32 participants while they watch 40 music video clips. All participants were students, consisting of 16 males and 16 females, with ages ranging from 19 to 37 years and an average age of 26.9 years. They were all in good physical health, had no mental disorders or brain injuries, and were right-handed. These stimuli are carefully selected to evoke various emotions, which provides a rich dataset for researchers to study the relationship between physiological responses, particularly EEG signals. 63 s of EEG signal data were collected by 32-channel Biosemi ActiveTwo device according to the international 10–20 system. The first 3 s of signals were the pre-trial baseline signals, and the remaining 60 s of EEG recordings were mood signals. At the end of each experiment, the subjects performed a self-assessment of their arousal and valence to determine whether the corresponding emotions were accurately elicited. The EEG signals were downsampled to 128 Hz. The electromyography (EMG) and electroencephalogram (EOG) signals were removed before the dataset was released. They were passed through a bandpass filter between 4 and 45 Hz to filter out noise. Emotions were classified into 4 categories based on the degree of arousal and level of valence (1–9) of each marker: happy, sad, fearful, and neutral.
- (2) The SEED dataset³⁴ is a large-scale open-source dataset developed by BCMI Labs for EEG-based emotion classification tasks, which can be divided into 3 categories of emotions (positive, neutral, and negative). Emotion clips were meticulously selected, ensuring that the clips can reliably evoke the corresponding emotions in subjects. The dataset consists of 15 emotion clips extracted from different Chinese films, each clip is approximately 4 min long and exclusively depicts one type of emotion. Fifteen healthy native Chinese students, including seven males and eight females, with an average age of 23, participated in the EEG signal acquisition experiment. EEG signals were recorded using a 62-channel ESI NeuroScan system with a sample rate of 1000 Hz. An experiment consisted of fifteen trials with each trial corresponding to an emotion film clip. Before each clip, there was a 5-s hint, followed by a 45-s self-assessment period, and a 15-s rest after each clip. After the experiment, based on the subjects' responses, only the sessions in which the target emotion was successfully elicited were selected for further analysis. The experiment was repeated three times for each subject. After the three sessions, all sessions for each subject were combined. EEG signals heavily contaminated by EMG and EOG were manually removed before the dataset was released. For data processing, the original EEG signals were first down-sampled to 200 Hz and then filtered to 4–50 Hz.
- (3) The SEED-IV dataset³⁵ is an evolution of the original SEED dataset, which is a multimodal dataset that include 62-channel EEG signals from 15 subjects, including eight females. Each subject watch 24 movie clips with a duration of approximately 2 min, introducing a change in the number of emotion categories from three to four: happy, sad, fear, and neutral. In addition to EEG signals, SEED-IV provides eye movement features recorded by SMI eye-tracking glasses. This multimodal dataset enhances its utility for emotion recognition by incorporating both EEG and eye movement data. For data processing, the original EEG signals were first down-sampled to 200 Hz and then filtered to 4–50 Hz.

Experimental settings

All experiments were conducted on the Dell 5810 server with the following main parameters: Intel Xeon E5-2660 processor, 64G of RAM, NVIDIA 2080Ti graphics card, and Centos 7.0 operating system. The Adam algorithm is used for training, the cross-entropy is employed as the loss function, and the entire model and recognition process were implemented using PyTorch framework.

For training the model, Adam was used with a batch size of 20 for all three datasets. The configuration for the learning rate, number of epochs, and weight decay varied for each dataset. DEAP dataset: Learning rate of $1e-5$, 300 epochs, and weight decay of 0.01. SEED and SEED_IV datasets: Learning rate of $1e-4$, 200 epochs, and weight decay of $1e-4$.

Evaluation metrics

In this experiment, three metrics—Accuracy (ACC), Standard Deviation (STD), and Average Time (AT)—are employed to evaluate the effectiveness of the proposed method. These metrics serve different purposes in assessing the model's performance. Accuracy (ACC): ACC is the ratio of correctly recognized EEG types, which can provide an overall measure of the model's correctness in recognizing emotion states. Standard Deviation (STD): STD reflects the dispersion of recognition accuracy across all epochs. It indicates the variability or consistency of the model's performance. Average Time (AT): AT represents the average time taken to identify each EEG emotion sample. This metric assesses the efficiency of the proposed method in processing and classifying emotion states. The definitions of these metrics are given as follows.

$$ACC = \frac{TP + TN}{N} \tag{15}$$

$$STD = \sqrt{\frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N}} \tag{16}$$

$$AT = \frac{T}{N} \tag{17}$$

where TP represents the true positive, indicating the number of correctly predicted EEG signals classified as positive cases, and TN represents the true negative, indicating the number of correctly predicted EEG signals classified as negative cases. *N* is the total number of EEG test samples. *x* is the achieved classification accuracy for each test experiment, and $\bar{x}$ is the average classification accuracy of all test experiments. *T* represents the time to identify the emotion categories of all EEG test samples.

Accuracy analysis

Accuracy (ACC) and Standard Deviation (STD) were used to evaluate the accuracy of emotion recognition and compare the performance of our method against existing approaches. The experiment is divided into two segments: (1) Assessment of Various Feature Selections with Transformer Encoder. This sub-section involves evaluating different feature selections when combined with the transformer encoder. The goal is to understand the impact of different features on the performance of the method. (2) Comparison with Existing Methods. The second sub-section focuses on comparing the performance of the proposed method with existing approaches. This step aims to provide insights into how well the proposed method performs relative to established methods in the field. In addition, a t-test is a statistical hypothesis test used to determine if there is a significant difference between the means of two groups. It is a simple yet powerful tool for comparing the means of two methods. By selecting the appropriate test type, calculating the t-statistic, and interpreting the p-value, we can objectively evaluate whether the observed differences between methods are significant. In our experiment, the p-value between the proposed MSDCGTNet and other methods was computed to assess whether the observed differences are due to chance or represent a true distinction between the two approaches.

Since the EEG signal is a type of multi-channel long time series data, most methods for emotion recognition using EEG signal rely on time-frequency conversion. This involves transforming the EEG signal into various spectrograms, such as wavelets^{36–38}, spectrograms and FFT (Fourier transform). Subsequently, multiple features are extracted from these spectrograms and fed into transformer encoder for emotion recognition. Table 1 presents the classification accuracy achieved through the combination of various time-frequency transformations with transformer encoders. It can be seen from Table 1 that all the methods demonstrated satisfactory performance since these methods uses local and global features. However, our proposed approach yielded the best results, because it used 1DCNN to extract features directly from the raw EEG signal. This not only reduces the computation cost but also avoids information loss due to the time-frequency conversion.

We also compared the proposed method to emotion recognition with other methods. All methods have been repeated, tested using the same dataset and evaluated using the same assessment indicators. Table 2 presents the recognition accuracy of various methods. References^{39,40} utilized the feature selection and SVM classifier for the purpose of emotion recognition from EEG signals, which can achieve accuracies of 92.3%, 83.99% and 70.58% across the DEAP, SEED and SEED_IV datasets, respectively. However, the artificial feature is subjective, so it tends to impact the accuracy of emotion recognition. Reference⁴¹ adopted 2DCNN to generate a virtual image from EEG signals for emotion recognition, and reference⁴² adopted two-level 1DCNN ensemble to predict the emotion states, which had the better detection accuracy, reaching 81.51% and 98.43% on DEAP datasets. However, they do not consider temporal features. GMSS⁴³ utilized graph-based multi-task self-supervised learning model for EEG emotion recognition, which achieved accuracies of 86.52% and 86.37% on the SEED and SEED_IV datasets, and reference⁴⁴ used DGCNN to classify graphs encoded with spatiotemporal information of the emotion EEG source, which achieve accuracy of 97.66% on the DEAP datasets, respectively, this was attributed to its ability to capture inherent spatial connections between distinct areas of the brain. However, they did not take into account temporal features. Reference⁴⁵ utilized recurrent neural networks (RNN) to handle emotion recognition, and references⁴⁶ used CNN and LSTM to detect the emotion, which reached

	DEAP		SEED		SEED_IV	
	ACC ± STD (%)	Time (s)	ACC ± STD (%)	Time (s)	ACC ± STD (%)	Time (s)
Spectrogram	94.25 ± 0.08 (<i>p</i> = 4e-3)	0.02	77.95 ± 0.24 (<i>p</i> = 4e-6)	0.05	95.17 ± 0.13 (<i>p</i> = 6e-3)	0.08
Wavelet transform	98.96 ± 0.01 (<i>p</i> = 3e-2)	0.09	98.12 ± 0.15 (<i>p</i> = 1e-2)	0.09	79.50 ± 0.18 (<i>p</i> = 7e-6)	0.15
FFT	99.56 ± 0.12 (<i>p</i> = 4e-2)	0.04	79.60 ± 0.99 (<i>p</i> = 4e-7)	0.04	91.03 ± 0.42 (<i>p</i> = 1e-4)	0.06
Our	99.66 ± 0.15	0.0043	98.85 ± 0.81	0.0052	99.67 ± 0.12	0.0077

Table 1. Comparison of efficiency and accuracy of DEAP, SEED and SEED_IV on different signal processing methods.

	DEAP				SEED		SEED_IV	
	Valence		Arousal		ACC ± STD (%)	Classes	ACC ± STD (%)	Classes
	ACC ± STD (%)	Classes	ACC ± STD (%)	Classes				
39	92.3 ± 0.12 ($p = 4e-5$)	2	86.3 ± 0.3 ($p = 1e-6$)	2	83.99 ± 11.72 ($p = 3e-2$)	3	70.58 ± 17.01 ($p = 6e-3$)	4
40	85.71/multimodal ($p = 1e-5$)				68.26 ($p = 6e-5$)	3	73.14 ($p = 6e-5$)	4
41	81.51 ($p = 3e-2$)	2	79.42 ($p = 3e-2$)	2	69.37 ± 0.20 ($p = 1e-4$)	3	68.96 ($p = 2e-4$)	4
42	98.43 ($p = 1e-5$)	2	97.65 ($p = 3e-5$)	2	87.33 ± 0.76 ($p = 1e-2$)	3	95.76 ($p = 6e-3$)	4
43	88.91 ± 0.05 ($p = 7e-5$)	2	92.73 ± 0.04 ($p = 5e-4$)	2	86.52 ± 6.22 ($p = 3e-3$)	3	86.37 ± 11.45 ($p = 3e-5$)	4
44	97.66 ± 0.02 ($p = 3e-3$)	2	97.78 ± 0.11 ($p = 7e-3$)	2	96.53 ± 0.54 ($p = 8e-3$)	3	86.24 ± 0.40 ($p = 2e-2$)	4
45	92.26 ± 0.07 ($p = 2e-5$)	2	86.33 ± 0.30 ($p = 6e-6$)	2	94.24 ± 5.95 ($p = 4e-2$)	3	79.37 ± 17.01 ($p = 4e-2$)	4
46	96.12 ± 1.65 ($p = 4e-3$)	2	96.66 ± 2.49 ($p = 2e-2$)	2	95.44 ± 3.32 ($p = 5e-4$)	3	91.67 ± 1.32 ($p = 6e-4$)	4
26	99.4 ($p = 4e-2$)	2	99.1 ($p = 1e-2$)	2	85.77 ± 0.83 ($p = 1e-5$)	3	98.78 ± 1.1 ($p = 4e-2$)	4
Ours	99.66 ± 0.02/4				98.85 ± 0.81	3	99.67 ± 0.12	4

Table 2. Performance comparison of DEAP, SEED, and SEED_IV data sets on different method.

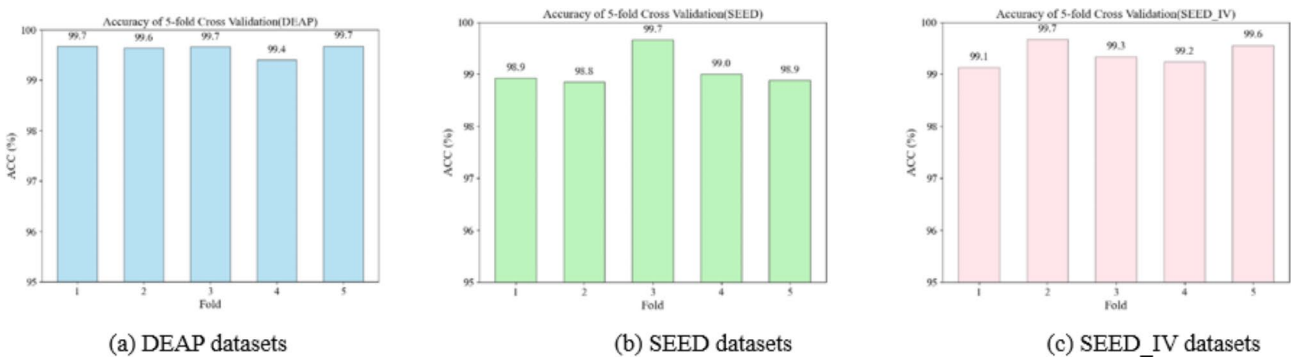


Fig. 7. Results of the 5-fold cross-validation.

accuracies of 94.24% and 95.44% on the SEED dataset, it is attributed to the effectiveness of RNN and LSTM in capturing temporal information. Nevertheless, this type of method requires a large amount of computation. Reference²⁶ used time-frequency conversion and transformer encoder for EEG emotion recognition, which had the higher detection accuracy, reaching 99.4% and 99.1% on the DEAP dataset. However, Time-frequency conversion results in information loss. Compared with the models in Table 2, our model presented the best performance among all candidate models, as: ① 1D-CNN extracted the local feature information, which avoids the information loss; ② Transformer Encoder extracted the global and temporal information, which ensures that the extracted information is comprehensive.

As shown in Tables 1 and 2, all p-values are less than 0.05, indicating that the accuracy improvement of the proposed method is statistically significant rather than a result of random chance. Therefore, compared to existing methods, the proposed method demonstrates a meaningful and substantial improvement in accuracy.

Given the limited amount of data and the uneven distribution of categories, 5-fold cross-validation was employed to ensure the model's robustness and its ability to generalize to new, unseen data. This method provides a more reliable estimate of the model's performance. Each dataset was randomly divided into five equally sized subsets, or 'folds.' For each fold, the model was trained on a combination of four folds (80% of the data) and validated on the remaining fold (20% of the data). This process was repeated five times, with each fold used exactly once as the validation set. Figure 7 shows the cross-validation results, demonstrating the effectiveness and practicality of this approach across different data splits. The experimental results indicate that the model consistently achieves high accuracy across all folds, highlighting its stability and strong generalization ability. These findings emphasize the method's robustness and effectiveness across diverse data partitions.

Efficiency analysis

Detection time is crucial for emotion recognition, as it determines whether the emotion recognition is real-time. Table 3 illustrates the run times of different methods on various datasets for EEG-based emotion recognition, and these details are utilized to demonstrate the efficiency of the proposed method.

As shown in Table 3, the recognition time of reference³⁹ and reference⁴⁰ are 1.34 and 0.09 s, respectively. The reason for the low recognition efficiency of the two methods is that the SVM cannot be accelerated via GPU. The recognition time of reference⁴¹ and reference⁴² are 0.04 and 0.12 s, respectively. The CNN method used in reference⁴¹ and reference⁴² can make use of GPU acceleration, so the detection efficiency is improved. Reference⁴³ employs GNN for emotion detection, which achieved the recognition time of 0.17 s, and reference⁴⁴ adopted

	DEAP				SEED		SEED_IV	
	Valence		Arousal		Total time (s)	Single time (s)	Total time (s)	Single time (s)
	Total time (s)	Single time (s)	Total time (s)	Single time (s)				
39	80.40	1.34	74.40	1.24	512.33	2.13	432.21	1.80
40	5.50	0.09	5.50	0.09	35.66	0.15	31.80	0.13
41	2.64	0.04	2.79	0.05	100.80	0.04	10.78	0.04
42	7.06	0.12	6.56	0.11	182.40	0.76	33.60	0.14
43	10.38	0.17	10.50	0.18	138.31	0.58	136.03	0.57
44	8.39	0.14	10.22	0.17	153.21	0.64	143.59	0.60
45	0.53	0.01	0.33	0.01	5.21	0.02	13.22	0.06
46	1.80	0.03	1.88	0.03	7.86	0.03	10.20	0.04
26	0.12	0.0020	0.12	0.0020	0.96	0.0040	2.16	0.0090
Ours	0.26	0.0043	0.26	0.0043	1.25	0.0052	1.85	0.0077

Table 3. Time efficiency comparison of different methods on DEAP, SEED and SEED_IV datasets.

Dataset	Parallel convolution branches (L)	ACC/%	Kappa	Model parameters/M
DEAP	1	93.59	0.8340	20.5
	2	98.30	0.9064	25.1
	3	99.59	0.9643	30.5
	4	99.66	0.9702	35.2
	5	99.97	0.9910	41.3
SEED	1	93.59	0.8202	29.3
	2	97.04	0.8293	33.9
	3	98.85	0.9398	38.3
	4	98.90	0.9492	42.8
	5	99.02	0.9602	47.3
SEED_IV	1	97.04	0.8905	22.4
	2	98.85	0.9342	30.1
	3	99.67	0.9786	38.5
	4	99.72	0.9853	46.5
	5	99.89	0.9943	54.5

Table 4. The model performance with different scale.

DGCNN to classify emotions, with the detection time set to 0.14 s. However, the utilization of high-order distant neighbors in GNN introduces a challenge known as “neighborhood explosion”. This occurrence demands more memory to store the exponentially increasing number of neighbor nodes, and significantly affects the model’s running speed. Reference^{45,46} use RNN or LSTM to detect the emotion state, which reached the recognition time of 0.01 and 0.03 s, respectively, but RNN and LSTM calculations depend on the outputs of previous time steps, which makes parallelization challenging and results in relatively slow operations. Compared with other methods, the proposed method in our paper and reference²⁶ can achieve the highest detection efficiency since transformer encoder have strong parallel processing capabilities, which can accelerate computation. The reason is that the latter requires a time-frequency transformation, thus it consumes more computation. Taking detection performance and flexibility into consideration, our method is very competitive for practical applications.

Ablation experiment

Ablation studies involve systematically analyzing the impact of removing or modifying specific components or features within a model. Table 4 shows the relationship between the number of convolutional scale and the performance of the approach. Table 5 illustrates the impact of the GLU and TCN modules on the performance of the proposed algorithm.

Table 4 shows that the classification accuracy increases as the number of parallel convolution branches increases. However, the number of model parameters also rises with the addition of more convolution branches. This indicates that a greater number of convolution branches allow the model to extract more enriched time-frequency domain features, thereby enhancing classification accuracy. While convolution operations are designed to extract hidden features, an excessive number of convolution kernels can increase computational load. Therefore, to achieve desired classification accuracy while maintaining efficiency, the network structure should remain as compact as possible. In summary, to improve network efficiency and reduce the number of parameters, this paper uses a multi-scale convolution with three branches to effectively extract and integrate EEG signal features.

Dataset	Model	ACC/%	P-value
DEAP	No GLU	98.04 ± 0.22	$3e-2$
	No TCN	94.34 ± 0.14	$2e-3$
	Ours	99.66 ± 0.15	–
SEED	No GLU	87.34 ± 0.43	$4e-5$
	No TCN	84.23 ± 1.73	$3e-5$
	Ours	98.85 ± 0.81	–
SEED_IV	No GLU	97.22 ± 0.32	$2e-2$
	No TCN	95.23 ± 1.21	$1e-3$
	Ours	99.67 ± 0.12	–

Table 5. The performance with different model.

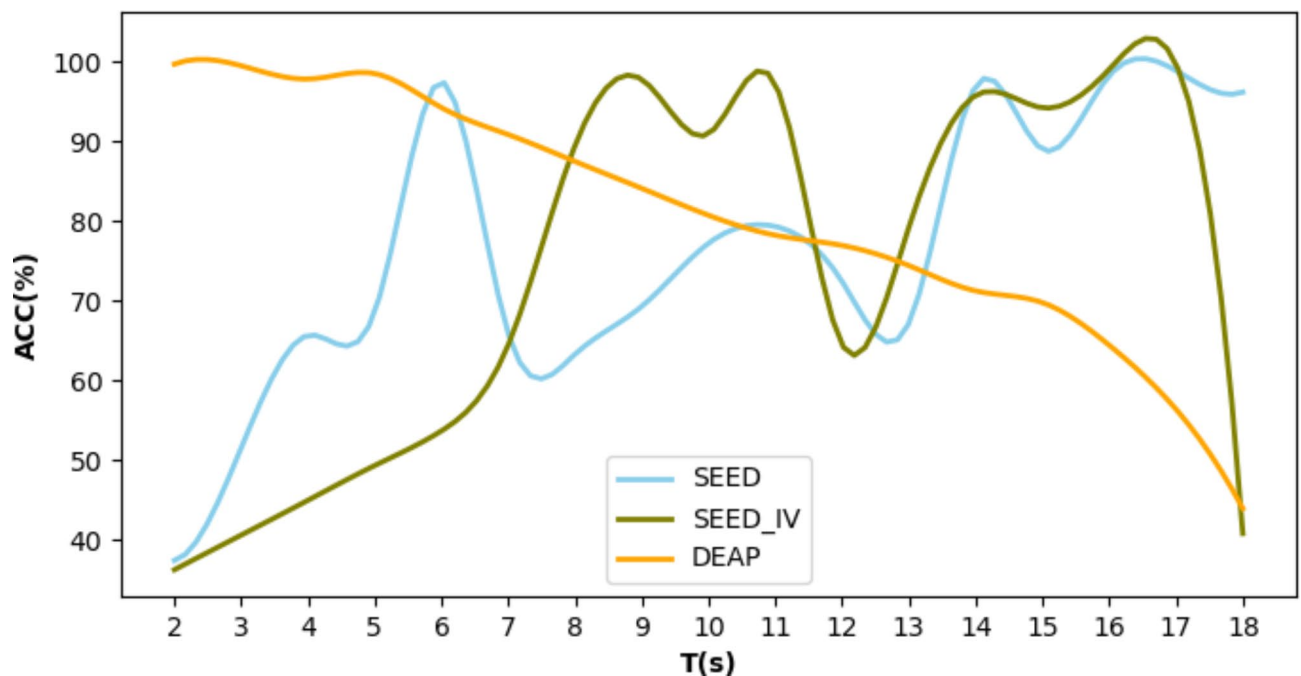


Fig. 8. Performances (average ACC $\pm$ STD %) of segment length on DEAP, SEED and SEED_IV.

Table 5 shows the performance of the models after each component was removed, in terms of accuracy and statistical significance (P-value). It can be seen that our model achieves the highest accuracy when both the GLU and TCN architectures are retained. Specifically, removing either GLU or TCN leads to a decrease in accuracy, indicating that both modules are crucial for improving model performance. Moreover, we performed a statistical significance test and the P-values for the comparisons with other models were all below 0.05, indicating that the proposed model significantly outperformed the alternatives. Based on these results, we conclude that both the GLU and TCN architectures make important contributions to the model's performance, and their combination enhances the model's effectiveness on the task.

Parameter analysis

Parameter selection and analysis were employed to further evaluate our proposed method. This paper analyzes the impact of two parameters on the experimental results. (1) EEG Segment Length: The length of different EEG segments contains varying amounts of information, thereby influencing the accuracy of emotion detection (2) Learning Rate and Number of Iterations: These parameters affect the convergence of the model.

As shown in Fig. 8, we illustrate the relationship between the length of the EEG segments and detection accuracy. The yellow line shows the variation of detection accuracy with different EEG segment length on the DEAP dataset, which reveals a tendency for accuracy to decrease as the segment length increases. The best performance is observed at a length of 2 s, which can achieve an accuracy of 99.66% and a standard deviation of 0.15. The blue line shows the variation of detection accuracy with different EEG segment length on the SEED dataset, which demonstrates that the detection accuracy fluctuates with the time slice size. The best performance is observed when the segment length is T = 17 s, which can reach an accuracy (ACC) of 98.85% and a standard deviation (STD) of 0.81. The green line shows the variation of detection accuracy with different

EEG segment lengths on the SEED_IV dataset. The recognition accuracy reaches more than 90% at segment length of ^{8,11} and ^{15,17}, and a segment length of $T=17$ s delivers the highest accuracy, with an average accuracy (ACC) of 99.67% and a standard deviation (STD) of 0.12. The experimental results highlight the importance of selecting the appropriate segment length for emotion recognition. It is crucial to stay within the optimal range, as exceeding it can lead to a significant reduction in classification accuracy.

The Fig. 9 illustrates the loss curve in relation to the learning rate. Figure 9a indicates that training on the DEAP dataset converges at approximately 300 epochs, when loss curves close to the convergence point, lr-0.0001 had the most accurate performance. Figure 9b showed that the loss curves on the SEED dataset converge successfully after about 200 epochs of training, lr-0.0001 was the best performance. Figure 9c showed that the curves on the SEED_IV dataset converge successfully after 200 epochs of training, lr-0.0001 had the highest accuracy. All the loss curves of model tend to be gentle in the later training period and only fluctuate across a small range convergence at different learning rates.

Interpretability of the model

SHAP (Shapley Additive Explanations) is a method for interpreting complex deep learning models by assigning an importance value to each feature for a given prediction. These values help identify the contribution of individual features to the model's predictions. Figure 10 illustrates the significance of the top 20 most influential features (channels) to the model across the three datasets. The figure highlights that the regions contributing to emotion detection include FP1, FP2, F3, F4, F7, and F8. These regions belong to the frontal area of the brain, which is responsible for cognitive and emotional regulation^{47,48}. This alignment indicates that the proposed emotion detection model corresponds well with the brain's known working mechanisms, further supporting the model's validity and effectiveness.

In summary, the experimental results demonstrate that our proposed method achieves high accuracy and efficiency in emotion recognition. Specifically, the method delivers satisfactory results across different datasets, indicating better adaptability to various EEG acquisition devices. However, our work has two limitations: (1) The model's training and testing are conducted on a server, which limits its practicality for real-world applications. (2) The model relies on large training datasets and faces challenges in generalizing to data from different EEG devices, potentially hindering its applicability in scenarios with limited data availability. To address these issues, future work will explore strategies such as transfer learning⁴⁹, domain adaptation, and model lightweighting to enhance the method's applicability and efficiency.

Conclusion

In this paper, a novel End-to-End Emotion Recognition Method from EEG Signals Using Multi-Scale Dynamic CNN and Gated Transformer Encoder is proposed. First, Multi-Scale Dynamic CNN is used to extract the temporal-spatial features from raw EEG signals, which mitigates the information loss caused by time-frequency conversion but also greatly enhances the detection performance. Secondly, the gated transformer encoder is utilized to capture global dependencies of EEG signals, thereby improving prediction and generalization capabilities and reduces computational demands through parallel processing of multiple self-attention mechanisms. Then, the temporal convolution network is used to extract temporal features available for EEG signals. Finally, SoftMax is utilized for emotion state classification. The experimental results show satisfactory performance on three public datasets (DEAP, SEED, SEED_IV). The proposed approach holds great potential for diverse applications within brain-computer interface (BCI) systems. In the future work, we will focus on utilizing the proposed model for in-depth EEG signal analysis and its application in disease diagnosis and mental health detection.

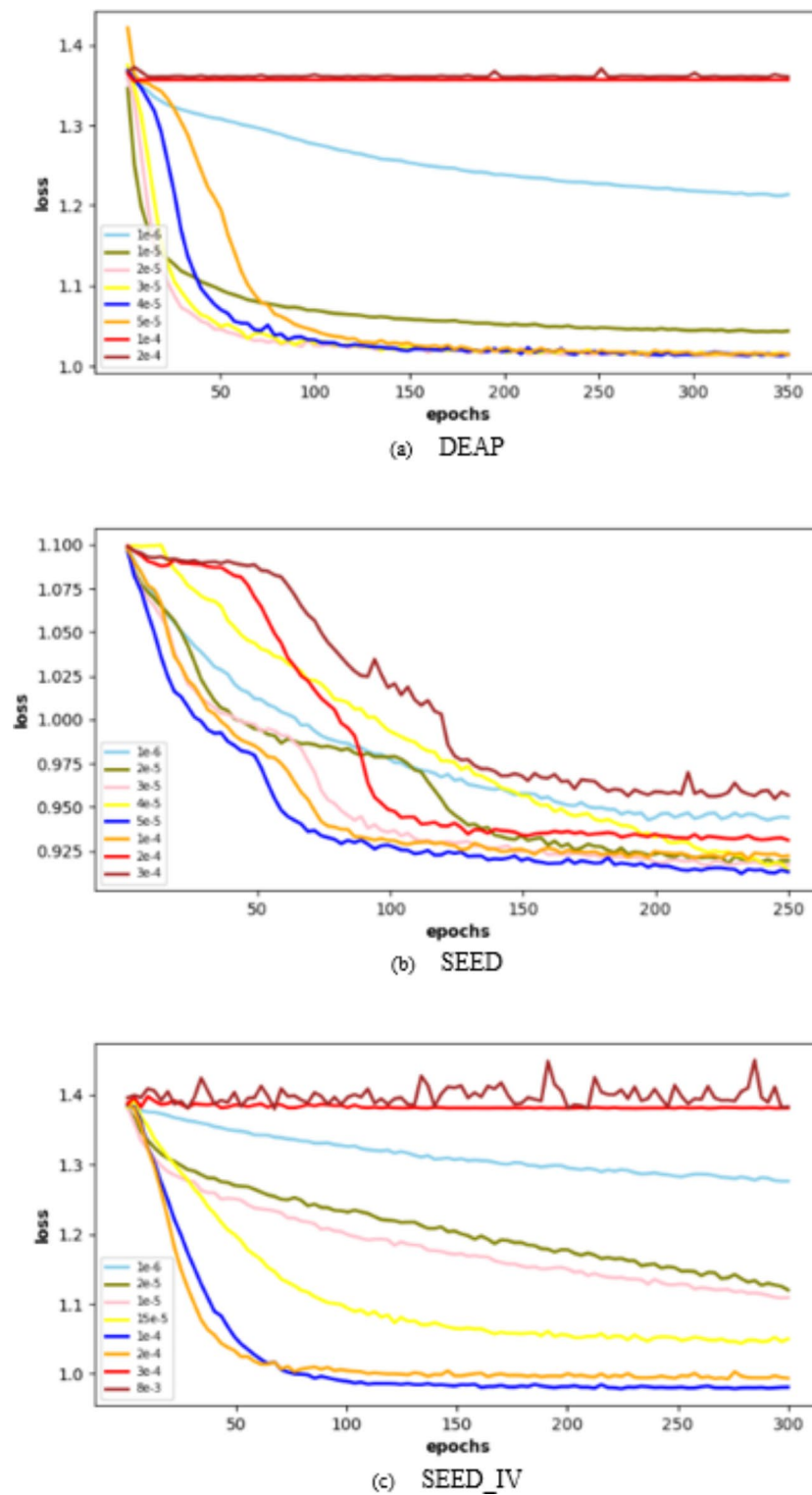


Fig. 9. The convergence at different learning rates.

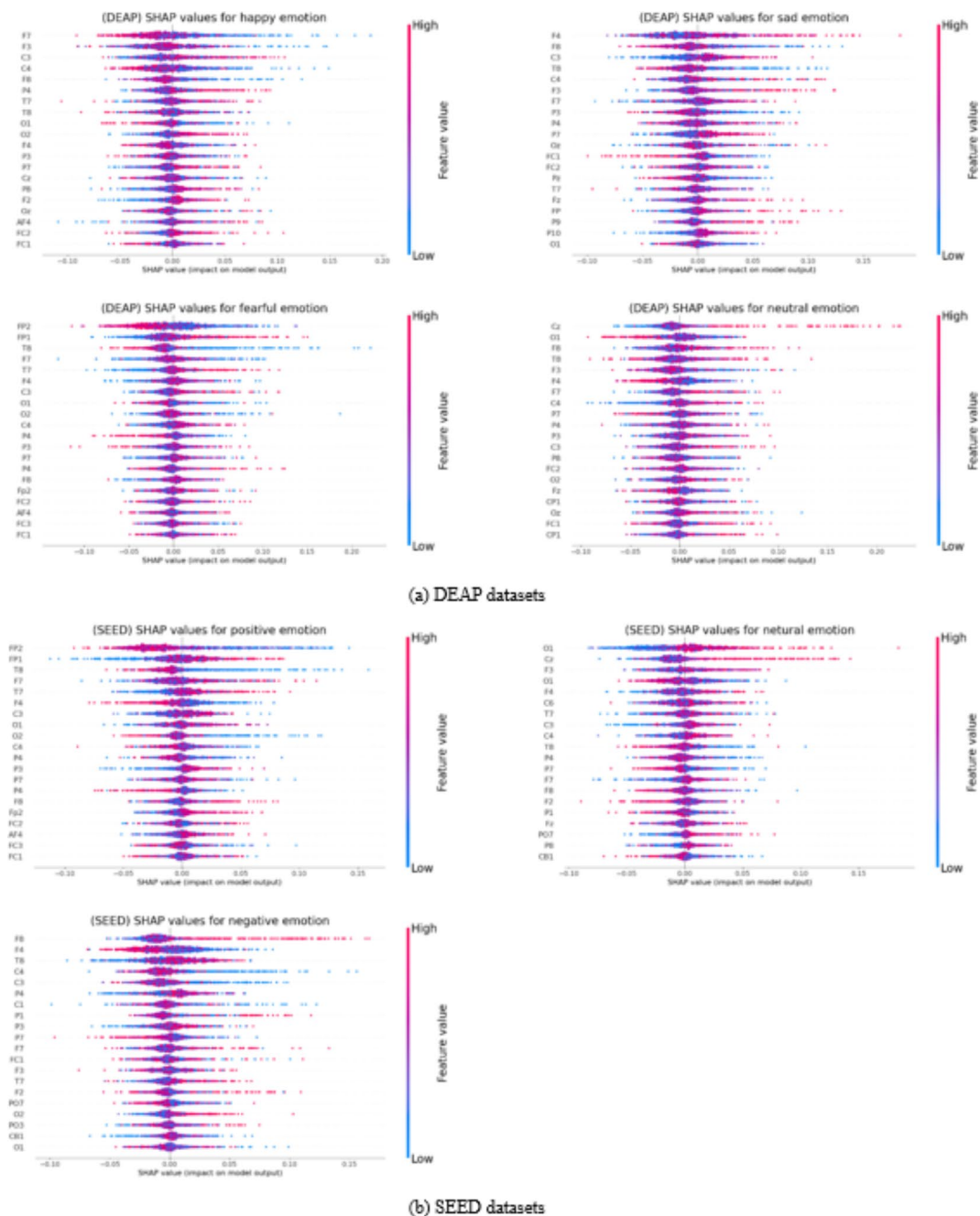


Fig. 10. SHAP value for different categories across three datasets.

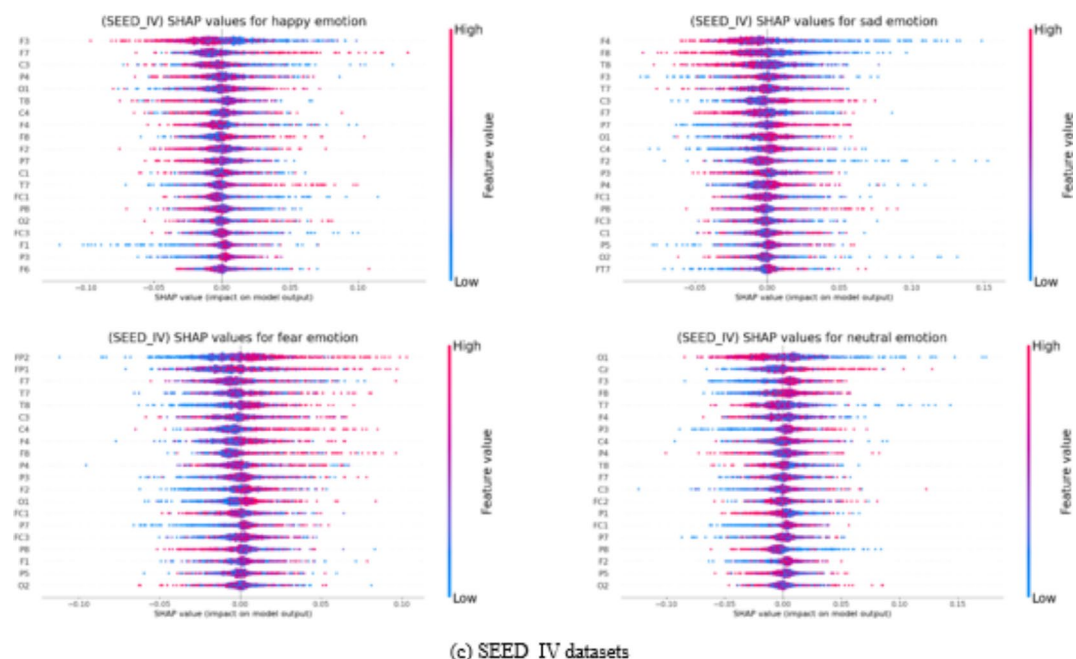


Figure 10. (continued)

Data availability

The dataset is shared open source. Availability Link: <https://www.eecs.qmul.ac.uk/mmv/datasets/deap/download.html>. <https://bcmi.sjtu.edu.cn/seed/seed.html>.

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Author contributions

Z.C. and X.K., data processing, experiments and evaluation, manuscript writing, Q.W., T.Y., J.T., supervision, conceptualization of the methodology, evaluation of the results, original manuscript and review. The authors confirm the absence of any participants in the article.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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